FEATURE FILM PRODUCTION // OCTOBER 2026

THE ART OF INVISIBLE VFX

An on-set supervision, 3D matchmove, and digital environment case study for feature film DEV1969.

01 ON-SET SUPERVISION

Live camera calibration, marker grids & director consultation.

02 SYNTHEYES 3D SOLVE

1,420 point tracking cloud solved with 0.38px residual error.

03 ENVIRONMENT INTEGRATION

Seamless volumetric cavern extension & lighting conform.

BENNY JOEL | ASSISTANT VFX SUPERVISOR & 3D TRACKING ARTIST

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PLATE DEV1969_CAVE_MATTE_V02 — FINAL CONFORM WITH VOLUMETRIC ATMOSPHERE

On-Set Camera Log & Field Survey

Soundstage calibration notes, lens distortion matrices, and HDR bracket passes.

SECTION 01: OPTICAL CALIBRATION & SENSOR TELEMETRY

Camera Body & Sensor	ARRI ALEXA Mini LF // 4.5K Open Gate (4448×3096)
Active Focal Length	Zeiss Master Prime 35.24mm (T1.3 at f/2.8)
Sensor Line-up / Mount	LPL Mount // Ground Glass Reticle 2.39:1 Anamorphic Ext.
Optical Height & Tilt	Nodal Center: 142.5 cm // Pitch: -8.4° // Roll: +0.2°
Timecode & Shutter	TC 14:22:08:12 // 180.0° Shutter (1/48s) // ISO 800

SECTION 02: PHOTOMETRIC GROUND TRUTH & SURVEY DATA

HDRI Dynamic Range	360° Spherical Panoramic // 7-Bracket RAW (1/8000s to 1/2s)
Chrome & Grey Spheres	Matched to practical stage tungsten & cool rim spill
Color Verification	X-Rite Macbeth ColorChecker v2.4 conformed to ACEScg
Stage Marker Matrix	50mm Fluo Green Crosses on calibrated 100cm floor grid
LiDAR Environment	FARO Focus S350 point cloud registered to millimeter scale

VFX SUPERVISOR FIELD VERIFICATION SIGN-OFF

I hereby verify that all camera parameters, tracking marker placements, and HDR bracket passes for Stage 03 were recorded under director supervision. Boundary cutoffs were negotiated to ensure zero actor overlap with digital cavern extension boundaries.

SUPERVISING ARTIST: BENNY JOEL — STATUS: APPROVED FOR MATCHMOVE



FIG 03. PRINCIPAL PHOTOGRAPHY // STAGE 03 LIVE BLOCKOUT (TAKE 04)

Live actors positioned within physical rock boundary; background cavern opening demarcated for seamless 3D set extension.

Live practical lighting: Warm 3200K tungsten spotlight on actor, cool ambient backlight to match moonlight.

CHAPTER II // PRODUCTION PIPELINE DECONSTRUCTION

From Soundstage Sensor to Final Cinema

Progressive breakdown of Shot 047 across raw capture, 3D matchmove, geometry alignment, and volumetric finishing.



01 RAW SENSOR ACQUISITION

ARRI ALEXA LF 4.5K Open Gate raw camera plate. Soundstage ceiling lights and floor tracking markers preserved for spatial registration.

LogC3 // Zeiss 35mm Prime // 24.00 fps



02 3D CAMERA TRACKING

Syntheyes Pro 2026 feature tracking solve. 1,420 multi-frame optical trackers calculated across 240 frames with 0.38px residual error.

Focal: 35.24mm // Residual: 0.38 px RMS



03 SPATIAL GEOMETRY FIT

Metahuman digital double & stasis chamber line-up. Used for physical shadow casting, occlusion, and cryogenic liquid refraction boundaries.

Quad-dominant mesh // Maya (.ma) Conform



04 VOLUMETRIC COMPOSITING

Unreal Engine 5 & DaVinci Resolve final conform. Cavern matte extension, dynamic light shafts, and photometric fog matching tungsten stage spill.

ACEScg // DCI-P3 4K // 100% Director Sign-off

STAGE 03 LIVE CAPTURE

4,448 × 3,096 RAW

ARRI Alexa Mini LF Open Gate calibrated sensor plate.

CAMERA TRACK ACCURACY

0.38 PX RESIDUAL

Sub-pixel reprojection error across 240 active cinematic frames.

CONFORM SIGN-OFF

100% DIRECTOR APPROVAL

Zero optical drift or edge jitter verified in DaVinci Resolve 4K QC.

FEATURE PRODUCTION: DEV1969 // SUPERVISED BY BENNY JOEL — ASST. VFX SUPERVISOR & TRACKING LEAD

CHAPTER III — MATHEMATICAL RECONSTRUCTION

Sub-Pixel Matchmove & Asset Fit

0.38 px
REPROJECTION RESIDUAL ERROR // 1,420 TRACKERS



SYNTHEYES 3D CAMERA SOLVE — CAMERA01 AT 24.00 FPS | 35.24MM SPHERICAL PRIME



METAHUMAN MATCHMOVE BUST

Used for exact spatial line-up, shadow casting, and collision boundaries with the stasis chamber cryogenic fluid.

- **Quad-dominant topology**
- **Real-world scale conform**
- **ARRI LF sensor match**

PRODUCTION PIPELINE SPECS

Primary Solver	Syntheyes Pro 2026 (Feature Track)
Sensor Gate	ARRI LF OpenGate (4448×3096)
Lens Distortion	3DEqualizer Anamorphic / Radial 4-order
DCC Hand-off	Autodesk Maya (.ma) + The Foundry Nuke (.nk)
Deliverable QC	DaVinci Resolve 10-bit Apple ProRes 4444

3D Level Design & Digital Background Plates

Modular subterranean cavern modeling, Lumen real-time global illumination, and volumetric depth matching for set extensions.



FIG 05A. UNREAL ENGINE 5 LEVEL DESIGN // MODULAR MASONRY & WATER

High-density Nanite masonry arches, procedural water reflection plane, and practical torchlight spill.

Unreal Engine 5.6 // Nanite Virtualized Geometry // Lumen Dynamic GI // 3200K Tungsten



FIG 05B. CONFORMED BACKGROUND PLATE // VOLUMETRIC EXTENSION PASS

Volumetric light shafts, moss-covered ceiling opening, and atmospheric haze matching stage depth.

ACEScg Color Space // Exponential Height Fog // 35.24mm Camera Frustum Lock // 100% Sign-Off

I. NANITE MODULAR KITBASHING

Constructed subterranean cavern architecture using modular high-density rock assets and stone arches, achieving cinematic density with zero LOD popping.

II. LUMEN REAL-TIME PHOTOMETRY

Configured multi-bounce indirect lighting to ensure digital cavern recesses naturally absorbed and scattered the warm 3200K tungsten spill of practical on-set lamps.

III. CAMERA FRUSTUM ALIGNMENT

Imported Syntheyes solved 3D camera curves directly into Unreal Sequencer, locking focal length (35.24mm) and optical nodal center for pixel-perfect plate alignment.

CHAPTER II — PRODUCTION CONFORM

Practical Reality vs. Digital Illusion

Direct comparison of on-set stage limitations and the final depth-matched 3D matte extension.



FIG 01. RAW CAMERA PLATE // OVERHEAD RIGS & MARKERS

Exposed soundstage ceiling, studio lights, and physical ground tracking tape requiring rotoscope & roto-paint.



FIG 02. FINAL COMPOSITED SHOT // VOLUMETRIC EXTENSION

Level design cave geometry, realistic light shafts, and depth-cued particulate haze matching live camera lens.

I. BOUNDARY SURVEY

Negotiated physical stage cutoff lines with the director on set to maximize practical actors while minimizing complex rotoscoping and clean-plate generation.

II. HDRI LIGHT CONFORM

Acquired 360° dynamic range brackets and chrome sphere passes to reproduce warm tungsten stage spill in digital shaders with matching photometric decay.

III. ZERO ARTIFACT SIGN-OFF

Delivered final 4K plate conformed to DaVinci Resolve grading workflow with 100% director sign-off and flawless sub-pixel edge convergence.